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# Entrepreneurial Orientation as a Determinant of Sustainable Performance in Polish Family and Non-Family Organizations

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**Abstract:** The paper investigates the relationship between entrepreneurial orientation and sustainable firm performance with the mediating role of innovation output in family and non-family businesses in the post-transition economy. In order to analyze these relationships, quantitative research on a sample of 322 Polish organizations was conducted. The research results show a few differences between family and non-family businesses concerning the influence of entrepreneurial orientation on innovativeness and sustainable organizational performance. Moreover, the study reveals the critical role of the proactiveness dimension in shaping long-term firm performance.

**Keywords:** entrepreneurial orientation; innovations; sustainable firm performance

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## 1. Introduction

The entrepreneurial orientation (EO) of companies refers to their "processes, practices, and decision-making activities that lead to new entry" ([1], p. 136) and, consequently, it reflects their "strategic posture towards entrepreneurship" ([2], p. 1579). The concept of EO [3], developed by Miller [4], is recognized as one of the most established concepts in management sciences [5]. Therefore, it has been a dominating concept in entrepreneurship research in recent years [6,7]. Although there have been various ideas regarding EO dimensions [8], the research instrument developed by Covin and Slevin [9] is most frequently used to measure entrepreneurship orientation [10]. Thus, the research commonly uses three key dimensions of EO [11]: innovativeness, risk-taking, and proactiveness. However, other approaches to the internal structure of EO are also present in the literature [12], and there are numerous proposals for the reconceptualization of the EO construct and discussions around it [2]. For example, Anderson, Kreiser, Kuratko, Hornsby, and Eshima (2015) proposed two other dimensions of EO. These cover entrepreneurial behavior (innovativeness and proactiveness) and managerial attitude towards risk.

According to the EO concept, enterprises can be more or less entrepreneurial. It depends on how much their management strives to innovate, how aggressively it enters new markets, and to what extent it accepts risk (its specific level) in the search and use of new opportunities [2]. Therefore, the EO is a precondition for the companies' competitive advantage in the contemporary turbulent and globalized environment. An important implication for companies is to shape EO to ensure innovations, the implementation of which will have a positive impact on the results [13,14]. Innovations are a significant manifestation of entrepreneurial behavior and at the same time are recognized as a factor improving companies' results and playing a unique role in creating a competitive advantage, and the survival of enterprises, especially in the long term [15].

Numerous studies investigated EO in different contexts [16]. They covered various determinants of EO, including psychological ones [17]. They also explored the impact of innovation (input and output) on firms' performances. Interestingly, there is little research considering the role of implemented innovations (innovation output) in the relationship

between EO and performance in family businesses (FBs) compared to non-family (NFBs) ones [18–21], although these two types of companies differ significantly in many fields. Furthermore, most research on these phenomena concerns the USA and highly developed Western countries [22,23]. Thus, there is a need to examine the relationship between EO and performance in FBs and NFBs in post-transitional economies. The present study seeks to fill this gap. It explores the relationships between EO and firm's performance in FBs and NFBs in Poland. This country transformed from a centrally planned economy to a market economy after 1989.

The main aim of the research presented in this paper was to determine if the relationship between EO, innovation output, and sustainable organizational performance is different for family and non-family businesses. In particular, this study focused on finding the answer to the following research question: "Are family and non-family businesses different when it comes to relationships between entrepreneurial orientation, innovation output, and sustainable firm performance?".

This study contributes to academic research and business practice. First, its results support the theory of FBs, as they reveal the differences between FBs and NFBs concerning the effect of proactivity on firm performance. Second, this study's outcomes enrich EO theory because they provide additional information about the key dimensions of EO. Moreover, by showing that proactiveness helps implement novel and valuable ideas, this study delivers arguments for FBs and NFBs' managers to strengthen their companies' proactiveness. Finally, this study develops the knowledge on post-transitional economies, as it was conducted in Poland, a country that, until 1989, was a centrally planned economy.

## 2. Literature Review

### *2.1. The Differences between Family and Non-Family Businesses*

*Family businesses* are firms where the owner finds the company as a family, family members own at least 50% of the shares, and at least one family member is involved in the company's management [24]. The family is understood as a specific social group related by marriage, biology, or adoption, including people also related by effect, commitment, dependence, and cooperation [25]. Among researchers of family enterprises, there is a justified belief that various aspects of the family affect management structures and processes, innovation, entrepreneurship, and financial efficiency [26].

Comparative analyses of various aspects of FBs and NFBs are carried out for two main reasons. Firstly, FBs hold an essential role in the economies of most countries [27], or even dominate the structure of their enterprises. It is estimated that FBs constitute 60% (Denmark) to even 98% (Cyprus) of all enterprises [28]. In Poland, FBs may constitute more than 70% of the small- and medium-sized enterprise (SME) sector. However, still the data on this issue vary significantly, depending on the research methodology (between 70% and 90%) [29]. Family businesses' positive impact on the economy's development, employment growth, and social welfare is widely appreciated [26].

Secondly, research confirms numerous differences between the two categories of enterprises [30], although the FBs also vary significantly [31]. The differences found between FBs and NFBs are, among other things, the innovation level, entrepreneurship, and entrepreneurial orientation (EO) [32]. The involvement of the founding family in the management of the company and its ownership, focusing on economic and non-economic goals (including handing over the company to the next generation), results in many differences between FBs and NFBs [33]. The presented research results on the typology of FBs allow for a better understanding of the differences between these enterprises and others (NFBs), especially in the field of entrepreneurship and innovation, which are determined by various combinations of factors related to both family and the characteristics of the environment or the involvement of non-family managers [31]. Some research results confirm that FBs are less innovative than NFBs [34]. Still, at the same time, other studies indicate that, generally, there are no significant differences in the level of innovation between FBs and NFBs [35]. Some articles reveal that FBs are more innovative than NFBs [14].

### *2.2. Entrepreneurial Orientation and Sustainable Performance in Family and Non-Family Firms*

As already mentioned, the entrepreneurial orientation (EO) of companies refers to their "processes, practices, and decision-making activities that lead to new entry" ([1], p. 136) and, consequently, it reflects their "strategic posture towards entrepreneurship" ([2], p. 1579). "Research examining the relationship between EO and business performance shows that EO leads to higher and more sustained firms' performance" [36,37]. Notably, that relation appears, although they use various measures, to be financial as well as non-financial, to capture enterprise performance. Among these measures, subjective measures of financial performance are most commonly employed. These include sales growth and return on investment (ROI), net profit in the last three years, or net profit in the last three years compared to major competitors [28].

Studies that have examined the EO–performance relationship in family businesses generate inconsistent results [12,38,39]. The positive influence of EO on family firm performance has been found by Hernandez-Perlines, Covin, and Ribeiro-Soriano [40]. Their study indicates that socioemotional wealth preservation concerns positively influence entrepreneurial orientation and family firm performance. Their findings also show that concern for socioemotional wealth preservation positively moderates entrepreneurial orientation's influence on family firm performance. Similarly, Lee and Chu [39] point out that EO positively impacts family firms' performance. Their study reveals that EO is potently associated with concurrent and sustaining performance when family ownership is combined with active family management and control. Nevertheless, that study also shows that the EO–performance relationship becomes insignificant in family firms where family governance is passive. The authors conclude that active family governance alleviates agency problems and facilitates stewardship within family firms. On the other side, a study by Arzubiaga et al. [41] shows that the link between entrepreneurial orientation and performance is more vital in family firms with lower family involvement levels. Furthermore, their findings highlight that this relationship's strength is growing in family businesses with higher levels of gender diversity on the board. The EO–performance relationship and family members' role in top management teams have also been explored by Calabrò et al. [42]. They found that identity-based solid faultiness negatively moderates the relationship between EO and family firms' performance even when the top management team comprises only family members. Moreover, they revealed that strong knowledge-based faultiness increases the positive impact of EO on family firms' performance.

Lumpkin, Brigham, and Moss [43], referring to the mixed results of previous research on EO and profitability in family businesses, highlight that the typical positive relationship between EO and companies' performance may be more complex in family than in non-family businesses. They state that family firms' long-term orientation is positively associated with innovativeness, proactiveness, and autonomy. However, it is negatively associated with risk-taking and competitive aggressiveness. It reveals that not all five dimensions of EO (innovativeness, proactiveness, risk-taking, competitive aggressiveness, and autonomy) are equally crucial for family firms' performance. Namely, only proactiveness, competitive aggressiveness, and autonomy are significant for family businesses.

Hence, the research results above indicate that it is still unclear whether the EO has the same or a diverse impact on the performance of FBs and NFBs. The fact is that family firms have their character [44]. They are based on family ownership, active participation of family members in company management [39,45,46], and multigenerational succession [47]. Thanks to the unity of ownership and management, agency problems are reduced there, and decisions can be made faster than in non-family companies. Therefore, family businesses can flexibly react to changes taking place in the market [39]. However, only some family firms are proactive in their environment [48]. Instead, many develop conservative innovative strategies [49] and are "reluctant to change and highly committed to their status quo" ([50], p. 312). By focusing on handing over the business to future generations, family businesses often avoid risky ventures [51] and, on average, are more risk-averse than non-family firms [52]. Such characteristics of family businesses may result in differences

between them and non-family firms in terms of EO, reflecting "the organizational processes, methods and styles that firms use to act entrepreneurially" ([1], p. 139). Thus, it can also affect the EO–performance relationship, when performance is considered in terms of company profits. Considering that, and based on the resource-based view (RBV) theory highlighting the role of idiosyncratic resources that are complex, intangible, and dynamic within a particular business unit in the behavior of a company [53], we assume that:

**H1.** *There are no significant differences between the impact of EO on the sustainable firm performance in family and non-family businesses.*

### *2.3. Innovation and Sustainable Performance in Family and Non-Family Businesses*

Innovation is a key economic development factor that plays an essential role in the competition in enterprises and economies [32]. It is critical in ensuring survival and dealing with competition by delivering value to customers/recipients [54,55]. Innovation can result in gaining more customers, improving a company's image, increasing sales and profits, or enabling entry into a new market. Research endeavors on innovation conducted so far mainly employ the approach which focuses on implementing new products. However, companies are sometimes mainly oriented towards this course of action. They may be interested in other types of innovation—for example, creating new business models or new markets [3].

According to the latest edition of the Oslo Manual [56], business innovation is defined as new or improved products or business processes (or a combination thereof) that differ significantly from previous products or business processes and that were introduced to the market or introduced to the company's operations ([56], p. 68). T Oslo Manual distinguishes two types of innovations: product innovation and business process innovation. The study of other types of innovations, going beyond product innovations, are mainly taken into account by the EO concept (its innovativeness dimension). It is essential not only because of recognizing and confirming the impact of other types of innovations (apart from product innovations) on business results, but it also confirms various manifestations of entrepreneurship [3].

In this research, innovation is treated as a dependent and independent variable—signifying its determinants and the results of the innovation activity observed in the organization. Innovations depend on many factors, both external [57] and internal [35]. However, in FBs, diverse factors affect innovation. The literature shows the role of familiness [42,58,59], family involvement [60–62], family ownership [63], and investments in innovation activities [60]. The literature presents the research results in positive [64] and negative relations between family involvement and expenditure on innovation in family enterprises [60,65].

It should be noted that the FBs and NFBs' innovation research considers both the innovation input and output approach [58,65]. Innovation is measured differently in research with the input innovation approach (here, mainly R & D expenditures) and differently with output innovation (here, mainly patents, number of innovations implemented) [66]. Research from the position of innovation input (R & D) in the context of the absorptive capacity performance of R & D confirms the differences between FBs and NFBs [67]. We have focused our attention on the innovation output. The differences in the level of innovation output in FBs and NFBs are confirmed by comparative studies on these categories of enterprises [32,35].

Most studies indicate a positive relationship between innovation and company performance [68,69]. Such results were found for large companies, as well as for small- and medium-sized enterprises [70]. At the same time, some studies show that these relationships may be context-dependent and, therefore, not always obvious [68,71].

To sum up, comparing the innovativeness of FBs and NFBs, there are divergent conclusions that family businesses are less innovative [72,73], or others emphasizing they are more innovative than their non-family counterparts [14]. Although they are unclear,

the research results on innovation in FBs confirm the family's multidirectional influence on innovation processes which affects innovation output [74].

When analyzing the relationship between innovation and sustainable performance in FBs and NFBs, it should be emphasized that these differences are not only a consequence of family involvement in the enterprise, but relate to the structure, strategy, goals, and resources of these entities [75].

**H2.** *Innovation output influence on sustainable firm performance is stronger in family firms, compared to in non-family businesses.*

### *2.4. The Influence of the EO on Innovation Output in FBs and NFBs*

Oslo Manual [56] points out that innovations in the organizations result from a diverse set of factors. Innovative activity, which is supposed to lead to innovation outcomes, requires, in particular, formal top management behaviors consisting of "evaluating opportunities for introducing changes, for example, through the use of analysis, creativity, and problem-solving methods" ([56], p. 86). The research on EO and service innovation relationships confirmed the positive relationship between the two dimensions of EO—risk-taking and proactive activities and innovation—but the negative relationship between autonomy and innovation. In turn, Pérez-Luño, Wiklund, and Cabrera [76], examining the relationships between proactivity and risk-taking and the number of innovations a firm generates, confirmed that these relationships turned out to be positive only in the case of a high number of innovations generated by a firm. Moreover, some studies show that risk-taking and risk aversion [77] are positively related to FBs' innovations.

At the same time, attention should be paid to another aspect of the relationship between EO and implemented innovations, as mentioned by Covin and Wales [3]: not all innovations implemented in an enterprise have to be the result of strategic top management behaviors. Some innovations may result from innovative employees' behavior, the effect of intrapreneurship. On the other hand, the study of entrepreneurial companies characterized by highly innovative abilities does not always allow for fully measuring these abilities (their effect) because the effect can only be reflected in sustainable performance [3], but not necessarily in the number of implemented innovations.

While the influence of EO on innovation in non-family businesses has been highlighted in the literature [78], the case is more complicated in family businesses. Existing research using the EO construct confirms that the relationships between EO and innovations in FBs are complex and may depend on many structural and behavioral variables [41,57]. According to the research, the family nature of the company, including family involvement in management [13], negatively affects the relationship between EO and innovation. In particular, it hinders the ability of family SMEs to turn EO into innovations. Family-dominated boards tend to have less knowledge and experience and are less diversified; they also have less networking capacity [13], limiting their turn of EO into innovations. Family businesses are also limited in terms of professional knowledge. In addition, their lack of social capital limits the innovativeness of a family business [79]. Moreover, previous research conducted in various countries indicates that FBs strive to avoid the risk associated with innovation [74], limiting the scope of investing in new products or services.

It is vital that the different dimensions of EO may influence the implemented innovations in different ways [76], and these relationships are not fully recognized, especially when considering the diversity of innovations. A significant part of the research confirms the differentiation in the level of innovativeness of FBs and NFBs, which may be a consequence of, among other things, differences in the strategic behavior of FBs compared to NFBs. This is more prevalent as the research indicates a particular paradox: FBs implement less innovation based on their abilities and potential than non-family enterprises. However, not all research results are consistent [32].

**H3.** *EO influences innovation output more intensely in family businesses, compared to in non-family businesses.*

### *2.5. Innovations as a Mediator in the Relationship between EO and Firm Sustainable Performance*

The complexity of the relationship between EO and firm sustainable performance prompts a search for variables that moderate this relationship [80]. As indicated by Lumpkin and Dess [1], the set of variables that can shape the relationship between EO and performance is numerous and includes organizational factors (e.g., strategy, structure, sizes) and environmental factors (e.g., environmental hostility, dynamism, complexity).

Mediating variables [81], such as knowledge-based variables [82] or organizational learning [83], among others, are also the subject of research on the relationship between EO and performance. As our earlier discussion shows, we assume that there is a relationship between EO and innovation, measured by the number of new solutions implemented (innovation output). Potentially, each of the dimensions of EO-innovativeness, risk-taking, and proactivity can impact innovation output, although the relationship may vary in strength and direction. The mediating role of innovation between the two dimensions of EO—risk-taking and proactivity and performance—has been investigated in small firms, and the confirmation of such relationships has been obtained [84]. Other studies have found the partial mediation effects of learning orientation and innovativeness on the relationship between EO and firm performance [85].

Similarly, from the analyses cited above, innovation output remains tied to performance, including the amount of business innovation implemented. As noted, most studies confirm the positive nature of these relationships, and few studies on the relationship between innovation and business performance show the absence of such relationships [86]. It should be noted that the identification of the relationship between innovation and corporate performance is related to the methods used to measure them, both innovation [66] and corporate performance, as well as the time perspective of the effects. In addition, in our further study of the relationship between EO and performance, the status of the enterprise (family or non-family) is taken into account. As indicated in our article, the differences between FBs and NFBs may affect the diverse level of their innovation, as confirmed by the studies cited above. However, many contextual factors may influence the benefits of innovation and sustainable firms' performance [70]. To summarize the analyses presented so far, in which we have tried to show that EO affects innovation and innovation affects performance, we propose the following hypothesis:

**H4.** *Innovation output mediates the relationship between EO and sustainable firm performance in both FB and NFB.*

## 3. Materials and Methods

A summary of the hypotheses tested in this research can be found in Figure 1.

### *3.1. Sample*

To study differences in relationships between EO, innovation, and sustainable firm performance in family and non-family businesses, we conducted empirical research on 322 Polish organizations. The pairing process was performed as suggested by Amann and Jaussaud [87]. In addition, the pairing process was done during the sample selection (sending invitations to companies and performing interviews) to guarantee 1. the equal number of family and non-family companies in the final sample; 2. comparability of two samples; 3. guaranteeing that every family firm of a certain size has its similarly large non-family counterpart. The sample selection strategy was as follows. In the first step, from the database of 10,009 companies, market research company responsible for data gathering randomly selected 600 family and 600 non-family companies. Companies were further paired according to their size (micro, small, medium, large). Then, the company sent invitations to participate in the research to pairs of companies. From this sample, 251 completed questionnaires were obtained by interviewers. Since the sample size was still below 350 minimum sample size for detecting the effects of EO on firm performance (to estimate minimum sample size to detect effects we used a-priori sample size calculator for structural equation models created by M. Soper (https://www.danielsoper.com/statcalc/

calculator.aspx?id=89, accessed on 1 May 2019), the market research company collected another 123 filled-in questionnaires using an online survey; it gave us 374 responses in total. Further, during the response analysis, we excluded 52 incomplete or mismatched answers, leading to 161 pairs of family businesses and their non-family counterparts. The data were gathered between September 2019 and January 2020.

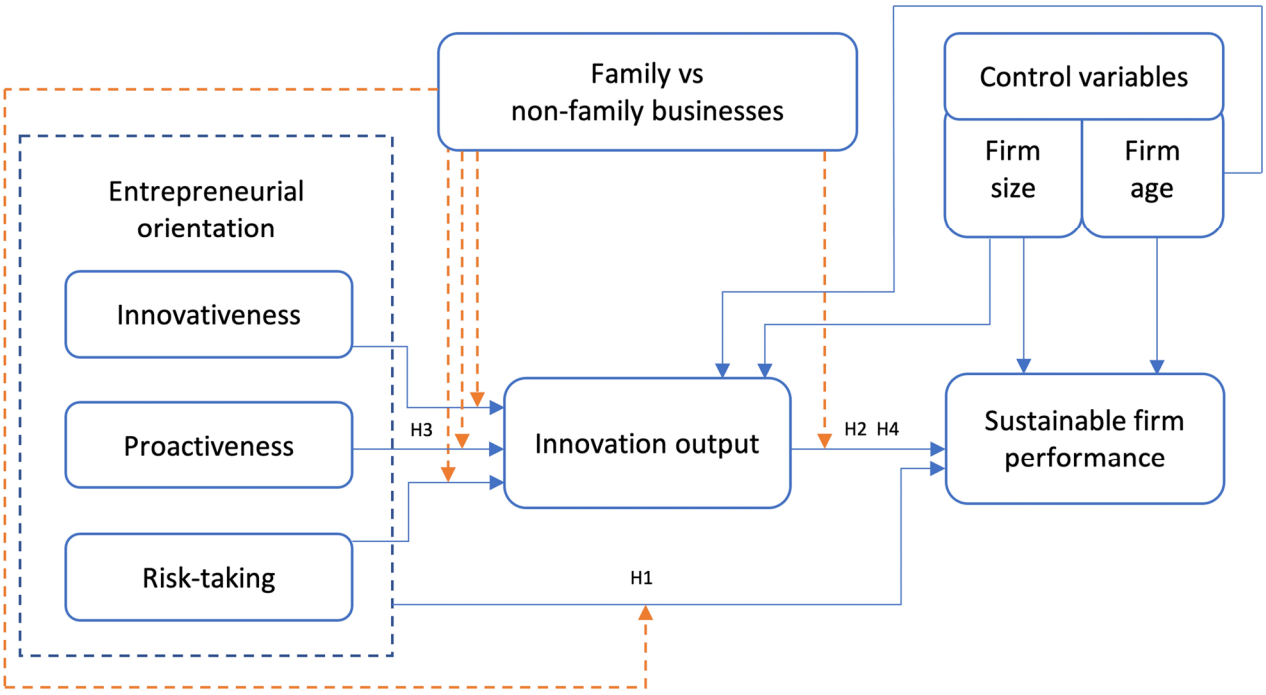



**Figure 1.** Research model. Relationships between variables are marked with solid blue lines; H4 hypothesis refers to mediation effect of innovation output in the relationship between entrepreneurial orientation and sustainable firm performance. Moderation effects are marked with dotted orange lines.

Finally, the sample was composed of 161 family and non-family business pairs. We understood a business as family-owned if: at least 50% of shares were in the control of family members, when the owner described a company as a family one, and when there was at least one family member engaged in the management of the company [24]. It is worth noting that in Poland, the FB category is more homogeneous than in other countries in terms of age (most were established in the last 30 years), in terms of size (most are micro and small companies), and in terms of ownership (the majority are 100% owned by one family) [88].

To test differences between two subsamples, we calculated the *t*-test for the differences in means assuming unequal variances of entrepreneurial orientation, firm performance, and innovation output, as measured by the number of innovations. The comparisons of data from the first and second round are given in Table 1.

As seen from the table above, data gained from the first and second rounds are different—in particular, means are lower than in the second round. We expect it to be due to the inherent characteristics of the studied companies and the composition (proportion) of family and non-family businesses in the two drawings. The ratio in the first round was 151 family firms and 72 non-family ones. In the second round, there were only 10 family firms selected for the research, and 89 non-family companies were researched.

In cases of family businesses, respondents were owners or high-level managers, and in referrals to non-family companies, they were either CEOs or high-level managers. Although the aim was to gather the data so that there would be an equal number of family and non-family businesses in terms of firm size and profile, this attempt failed in the data-gathering process. Finally, the sample structure is as follows (Table 2).

**Table 1.** The comparison of data gathered during first and second round.

| | Number of Questionnaires | Number of Questionnaires Excluded | Number of Questionnaires Retained | Mean and SD Value of EO | Mean and SD Value of Innovation Output | Mean and SD Value of Firm Performance |
|---|---|---|---|---|---|---|
| First round | 251 | 28 | 223 | 3.21 | 24.35 | 4.53 |
| Second round | 123 | 24 | 99 | 3.02 | 21.11 | 4.41 |
| Total | 374 | 52 | 322 | 3.15 | 23.36 | |
| Difference | | | | 0.19 | 3.24 | 0.12 |
| *t*-test statistic | | | | 2.030 | 1.477 | 1.096 |
| Ha diff < 0 | | | | Pr(T < t) = 0.9782 | Pr(T < t) = 0.9296 | Pr(T < t) = 0.8629 |
| Ha diff ! = 0 | | | | Pr(\|T\| > \|t\|) = 0.0435 | Pr(\|T\| > \|t\|) = 0.1408 | Pr(\|T\| > \|t\|) = 0.2743 |
| Ha diff > 0 | | | | Pr(T > t) = 0.0218 | Pr(T > t) = 0.0704 | Pr(T > t) = 0.1371 |

**Table 2.** Profile and size of companies included in the final sample.

| | Family | Non-Family | Total |
|---|---|---|---|
| | Company Profile | | |
| **Trade** | 18 | 22 | 40 |
| **Services** | 51 | 60 | 111 |
| **Production** | 46 | 23 | 69 |
| **Mixed** | 46 | 56 | 102 |
| **Total** | **161** | **161** | **322** |
| | Company Size | | |
| **Micro (1–9)** | 26 | 21 | 47 |
| **Small (10–49)** | 75 | 87 | 162 |
| **Medium (50–249)** | 43 | 34 | 77 |
| **Large (250+)** | 17 | 19 | 36 |
| **Total** | **161** | **161** | **322** |

### *3.2. Analysis Methodology*

The data analysis started with the computation of frequencies concerning family and non-family businesses in terms of company profile and size (Table 2 above). These were calculated in Stata for Mac. Then we proceeded with checking the reliability of scales using Cronbach's alpha coefficients [89] for constructs. In the following step, confirmatory factor analysis was carried out on sustainable firm performance and entrepreneurial orientation. These were computed in Mplus for Mac. For sustainable firm performance, we predicted, according to the scale use, a single factor structure of the construct [90]. For entrepreneurial orientation, considering previous suggestions by Kreiser, Marino, and Weaver [91], we tested three dimensions: innovativeness, proactiveness, and risk-taking. After confirming three-dimensional structure of the construct, Cronbach's alphas were calculated for each construct separately to check for internal consistency.

In the following step, Pearson correlations and descriptives were calculated between constructs. For this instance, metavariables were used, as suggested by Wójcik-Karpacz, Kraus, and Karpacz [92]. The analysis was carried out in Stata. This allowed for identification of relationships between constructs and allowed for further analysis. In the following steps, we checked for multicollinearity between constructs. To this purpose, we first calculated correlations, then regressed dependent variables on independent variables, and finally used the command VIF (all computed in Stata for Mac). Checking for Variance

Inflation Factors allows for the identification of multicollinearity between variables, which may prevent a further misinterpretation of results [93].

Further, structural equation models of relationships were calculated in Mplus for Mac, considering the internal structure of sustainable firm performance (single latent variable) and entrepreneurial orientation in three dimensions (innovativeness, proactiveness, and risk-taking). We started with a model composed of only sustainable firm performance (main dependent variable) and control variables: size and age of a company (decimal logarithms of the firm's size and age were used to standardize the data). Next, we added entrepreneurial orientation dimensions to the model (a main independent variable). Finally, innovation was added to the model as a mediator. In every case, analysis was performed following recommendations provided by Muthen and Muthen for multigroup latent structural equation modeling [94]. In practice, for each step we calculated two models: the first with all parameters constrained and the second with parameters unconstrained, considering that samples of family and non-family businesses were independent. To test the differences between models, using chi-square and degrees of freedom, right-tailed probability of the chi-squared distribution in MS Excel was calculated. This allowed for the comparison of models created for family and non-family companies [95].

### *3.3. Measures*

Sustainable firm performance was measured using a modified five-item scale originally suggested by Antoncic and Hisrich [90] (Cronbach's alpha = 0.8622). The studies recognized that comparisons with competitors are an important issue for evaluating effectiveness [8]. Therefore, the subjective indicators previously developed by Antoncic and Hisrich [90] were used, replacing the term "closed cafeterias" with a seven-point Likert scale to obtain correspondence to other measures. The scale was previously used in Polish conditions in this form and proved its validity and reliability. Respondents were asked how they compared with their main competitors in the last three years on average annual employment growth, average annual sales growth, market share measured by the volume (volume) of sales, market share measured by sales value, and net profit (sample item: "Average annual employment growth over the last three years compared to major competitors"). Since previous studies reported a single factor structure of the scale, we performed confirmatory factor analysis in mPlus 8.0. RMSEA equaled 0.041, and CFI and TLI achieved 0.996 and 0.992, respectively, which confirms a single factor structure.

Entrepreneurial orientation is frequently measured in three aspects [96]. However, some studies report five dimensions [1,97,98]. In this vein, Wales et al. [7] suggest enriching the concept with diverse combinations of EO dimensions. In our study, EO was measured using a scale proposed by Kreiser, Marino, and Weaver [91], which is an adaptation from the original Covin and Slevin scale [9,99] and was also tested in Polish conditions [100,101]. It consists of 8 questions related to three constructs: innovativeness, proactiveness, and risk-taking. The scale used a selection procedure on a continuum, the ends of which have detailed descriptions, and the answers were given using a five-point Likert scale. Innovativeness was measured using three items (sample question: "Senior management in the company in general: Emphasizes marketing of tried and tested products/services—Emphasizes research and development, innovation, development of new products/services"; Cronbach's alpha = 0.7388). Proactiveness was measured using 3 items (sample question: "When dealing with competitors, our company: responds to actions initiated by competitors—starts actions itself, which are later responded to by competitors"; Cronbach's alpha = 0.7432). Risk-taking was measured using 2 items (sample question: "Senior management in our company is generally in favor of: undertaking low-risk projects (with normal and certain profitability)—undertaking high-risk projects (with a chance of high profitability)"; Cronbach's alpha = 0.8072). Cronbach's alpha for the whole scale equaled 0.8804.

The innovation output (a mediator in our study) was measured following the Oslo Manual [56] recommendations as the number of innovations implemented within the last

three years in respect to products, processes, marketing, and organizational innovations (as a sum of innovations generated and implemented by a company) (sample question: "Has a new or improved product or service been implemented in the last 3 years that is significantly different from the company's previous product or service and has been introduced to the market? If so, please specify how much?"). In addition, we took into account several management innovations, defined as new management methods in the field of planning combined with the implementation of new organizational structures and incentive/remuneration/control systems [102]. We included the total number of innovations of all types for our analysis and calculated the logarithm to normalize the scale.

To avoid distorting the study's results by the influence of exogenous factors, we took into account two control variables—the age of the company and its size. We included the company's age, which is often defined as an important factor explaining the firm's performance [103]. The company's age is measured by the number of years from its foundation.

Firm size is often perceived as an important factor explaining the company's high efficiency [10] and innovative abilities [104]. Moreover, it is a control variable that has been taken into account in many previous studies testing the relationship between EO and sustainable firm performance [105]. Firm size was measured by the number of employees (full-time equivalents).

## 4. Results

To better understand our data, we firstly calculated the descriptive statistics and Pearson's correlations between research constructs (calculated as means). We noted that the logarithms were calculated for the number of employees (size of a company) and number of years in the market (age of a company) to bring them to scale (see Table 3).

**Table 3.** Descriptives and correlations.

| Variables | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
|---|---|---|---|---|---|---|---|
| (1) Sustainable firm performance | 1.000 | | | | | | |
| (2) Innovativeness | 0.325 * | 1.000 | | | | | |
| (3) Proactiveness | 0.500 * | 0.617 * | 1.000 | | | | |
| (4) Risk-taking | 0.400 * | 0.727 * | 0.649 * | 1.000 | | | |
| (5) Innovation output (log 10) | 0.393 * | 0.504 * | 0.479 * | 0.415 * | 1.000 | | |
| (6) Size of a company (lot 10) | 0.211 * | 0.193 * | 0.250 * | 0.161 * | 0.337 * | 1.000 | |
| (7) Age of a company (log 10) | −0.025 | 0.121 * | 0.097 | 0.117 * | 0.075 | 0.282 * | 1.000 |
| Mean | 4.498 | 3.254 | 2.981 | 3.269 | 1.197 | 1.541 | 1.223 |
| Std. Dev. | 0.916 | 0.92 | 0.896 | 1.058 | 0.424 | 0.55 | 0.275 |
| Min | 1.6 | 1 | 1 | 1 | 0 | 0.699 | 0.301 |
| Max | 7 | 5 | 5 | 5 | 2.079 | 2.845 | 2.079 |

* $p < 0.05$, $n = 322$.

The analysis of Table 3 leads us to conclude that firm performance is strongly connected to proactiveness, risk-taking, innovativeness, and innovation output. Correlation coefficients indicate there should be statistically significant relationships between independent variables and organizational performance. However, the direction of these relationships needs to be further examined. All three entrepreneurial orientation dimensions are significantly correlated. Furthermore, innovation output is related to all three dimensions of entrepreneurial orientation. The size of a company appears to correlate moderately with organizational performance and innovation output. It is also significantly related to all three entrepreneurial orientation dimensions. Finally, the age of a company correlates with innovativeness and risk-taking and covaries with the company's size.

The analysis of the correlation table allows for stating that there are significant relationships between individual constructs, but to gain a deeper understanding and verify hypotheses, further analyses are needed.

Moreover, the review of Table 3 reveals significant correlations between all three dimensions of entrepreneurial orientation (ranging from 0.617 to 0.727). Due to these high correlations, we decided to check for multicollinearity between variables. For this instance, we calculated variance inflation factors in Stata for Mac. The table with the results is presented below (Table 4).

**Table 4.** Variance Inflation Factors matrix.

| Variable | Sustainable Firm Performance | Innovativeness | Proactiveness | Risk-Taking | Innovation Output | Size |
|---|---|---|---|---|---|---|
| Innovativeness | 2.47 | | | | | |
| Proactiveness | 2.47 | 1.93 | | | | |
| Risk-taking | 2.01 | 1.79 | 1.33 | | | |
| Innovation output | 1.53 | 1.43 | 1.22 | 1.22 | | |
| Log Size | 1.24 | 1.24 | 1.22 | 1.13 | 1.09 | |
| Log Age | 1.10 | 1.10 | 1.10 | 1.09 | 1.09 | 1 |
| Mean VIF | 1.8 | 1.49 | 1.22 | 1.15 | 1.09 | 1 |

The analysis of variance inflation factors reveals a low probability of multicollinearity that would affect further analyses since VIF are significantly lower than 5. Thus, we decided to dig deeper into the data to analyze the relationships between all the constructs. We decided to employ structural equation model to gain a deeper understanding of the relationships and test hypotheses while verifying all the relationships at once, giving the researcher the ability to grasp the big picture of the studied associations. We decided to analyze the data by gradually increasing the complexity of the model—we calculated three research models. In the first one, we regressed sustainable firm performance (a latent variable) on the logarithm of size and the logarithm of the age of an organization. The second model is enriched with the influence of the entrepreneurial orientation dimensions: innovativeness, proactiveness, and risk-taking (all treated as reflective constructs—latent variables) on the sustainable firm performance (also analyzed as a reflective construct). Finally, model three accounts for the mediating effect of innovation output (measured as the decimal logarithm of a number of innovations generated in the recent three years) in the relationship between entrepreneurial orientation and firm sustainable performance. The model estimation results are provided in Table 5.

For clarity, below, the estimated models in family and non-family firms are presented in Figures 2 and 3, respectively.

Models were estimated using multigroup analysis in MPlus for Mac. In order to compare if there are significant differences between family and non-family companies (i.e., differences in the internal structure of constructs and differences in the relationship between constructs), every model was computed two times. In the first estimation, the so-called "restricted" model was calculated, which did not account for differences between family and non-family firms (respective relationships were set to be equal in both types of firms). Next, the relationships were "freed up," and the unrestricted model was calculated (it produced two separate models: one for family and one for non-family organizations). The restricted to unrestricted model is compared using the right-tailed probability of the chi-squared distribution in MS Excel by comparing the chi-squares and degrees of freedom of these two models. It should allow for answering the question of whether the differences between family and non-family businesses are significant enough to claim the differences in the relationships between these two types of organizations.

**Table 5.** Relationships between entrepreneurial orientation, organizational performance, and innovation output in family and non-family businesses.

| Variable | Model 1 (Overall) (Restricted) | Model 1 (Family) (Unrestricted) | Model 1 (Non-Family) (Unrestricted) | Model 2 (Overall) (Restricted) | Model 2 (Family) (Unrestricted) | Model 2 (Non-Family) (Unrestricted) | Model 3 (Overall) (Restricted) | Model 3 (Family) (Unrestricted) | Model 3 (Non-Family) (Unrestricted) |
|---|---|---|---|---|---|---|---|---|---|
| Fit Parameters | | | | | | | | | |
| CHI2 | 55.248 | 54.883 | | 318.582 | 313.618 | | 346.742 | 336.942 | |
| DF | 36 | 34 | | 189 | 184 | | 213 | 202 | |
| RMSEA | 0.058 | 0.062 | | 0.065 | 0.066 | | 0.062 | 0.064 | |
| CFI | 0.975 | 0.973 | | 0.937 | 0.937 | | 0.940 | 0.939 | |
| TLI | 0.972 | 0.968 | | 0.931 | 0.929 | | 0.933 | 0.928 | |
| Akaike Information Criteria (AIC) | 4258.589 | 4262.224 | | 11,060.227 | 11,065.263 | | 11,289.576 | 11,301.776 | |
| SRMR | 0.052 | 0.051 | | 0.077 | 0.075 | | 0.076 | 0.074 | |
| R2 (Performance) | 0.063 (0.030; 0.033) * | 0.048 (0.034; 0.166) | 0.082 (0.046; 0.074) | 0.408 (0.070; 0.000) * | 0.347 (0.080; 0.000) * | 0.618 (0.198; 0.002) * | 0.438 (0.071; 0.000) * | 0.383 (0.078; 0.000) * | 0.658 (0.251; 0.006) * |
| R2 (Innovativeness) | - | - | - | - | - | - | 0.409 (0.063; 0.000) * | 0.404 (0.069; 0.000) * | 0.513 (0.236; 0.030) * |
| Dependent variable: Organizational Performance | | | | | | | | | |
| Size | 0.345 (0.084; 0.000) * | 0.300 (0.114; 0.008) * | 0.395 (0.120; 0.001) * | 0.177 (0.075; 0.018) * | 0.113 (0.103; 0.249) | 0.242 (0.107; 0.023)* | 0.112 (0.077; 0.144) | 0.037 (0.104; 0.724) | 0.135 (0.159; 0.395) |
| Age | −0.304 (0.167; 0.068) | −0.251 (0.260; 0.334) | −0.356 (0.218; 0.103) | −0.339 (0.148; 0.023) * | −0.265 (0.230; 0.249) | −0.422 (0.194; 0.030)* | −0.321 (0.146; 0.028) * | −0.240 (0.224; 0.284) | −0.405 (0.197; 0.039) * |
| Innovativeness | - | - | - | −0.365 (0.246; 0.137) | −0.040 (0.250; 0.873) | −1.213 (1.148; 0.291) | −0.581 (0.284; 0.041) * | −0.178 (0.268; 0.506) | −1.535 (1.608; 0.340) |
| Proactiveness | - | - | - | 0.849 (0.214; 0.000) * | 0.602 (0.217; 0.006) * | 0.892 (0.501; 0.075) | 0.786 (0.213; 0.000) * | 0.547 (0.213; 0.010) * | 0.706 (0.708; 0.319) |
| Risk-Taking | - | - | - | 0.071 (0.202; 0.725) | 0.007 (0.179; 0.969) | 0.751 (1.149; 0.513) | 0.180 (0.209; 0.389) | 0.038 (0.177; 0.830) | 0.975 (1.530; 0.524) |
| Innovation output | - | - | - | - | - | - | 0.383 (0.153; 0.012) * | 0.431 (0.178; 0.016) * | 0.663 (0.751; 0.377) |
| Indirect effects: Mediation analysis | | | | | | | | | |
| Path: Innovativeness—Innovation output—Performance | - | - | - | - | - | - | 0.149 (0.089; 0.097) | 0.136 (0.088; 0.121) | 0.516 (0.947; 0.586) |
| Path: Proactiveness—Innovation output—Performance | - | - | - | - | - | - | 0.057 (0.035; 0.105) | 0.056 (0.046; 0.218) | 0.221 (0.393; 0.574) |
| Path: Risk-Taking—Innovation output—Performance | - | - | - | - | - | - | −0.059 (0.049; 0.226) | −0.030 (0.043; 0.481) | −0.418 (0.855; 0.625) |
| Dependent variable: Innovation output (in brackets S.E and *p*-values are given) | | | | | | | | | |
| Constant | - | - | - | - | - | - | 0.987 (0.101; 0.000) * | 0.985 (0.156; 0.000) * | 0.927 (0.155; 0.000) * |
| Size | - | - | - | - | - | - | 0.172 (0.038; 0.000) * | 0.177 (0.052; 0.001) * | 0.162 (0.054; 0.003) * |
| Age | - | - | - | - | - | - | −0.052 (0.074; 0.488) | −0.056 (0.116; 0.631) | −0.027 (0.098; 0.779) |
| Innovativeness | - | - | - | - | - | - | 0.388 (0.126; 0.002) * | 0.317 (0.140; 0.024) * | 0.779 (0.603; 0.197) |
| Proactiveness | - | - | - | - | - | - | 0.148 (0.090; 0.100) | 0.131 (0.105; 0.212) | 0.333 (0.329; 0.311) |
| Risk-Taking | - | - | - | - | - | - | −0.155 (0.102; 0.126) | −0.070 (0.096; 0.465) | −0.630 (0.653; 0.335) |

Relationships statistically significant are marked with * in brackets S.E. and *p*-values are given.

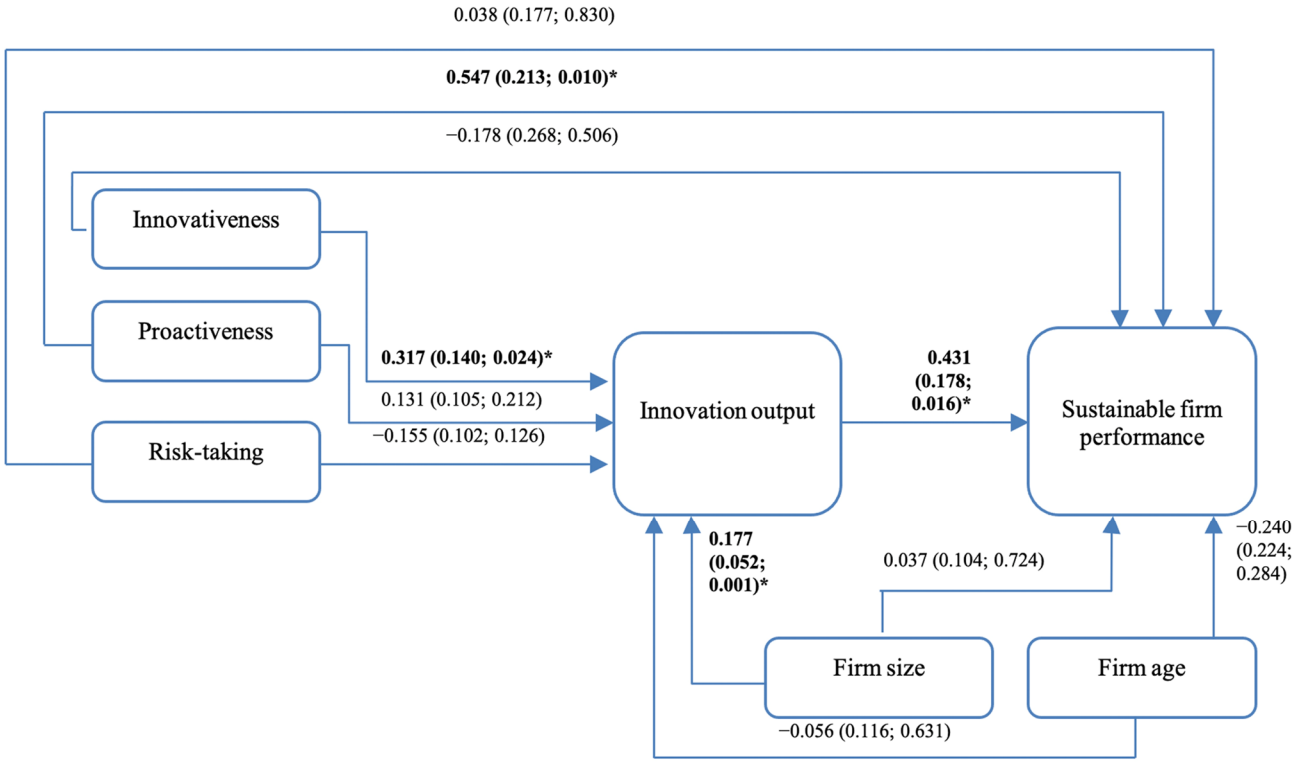



**Figure 2.** Relationships between entrepreneurial orientation, innovation output, and firm performance in family firms. Above arrows parameter estimates are given; in brackets standard errors and *p*-values are provided, bold font size and asterisk are used for statistically significant relationships.

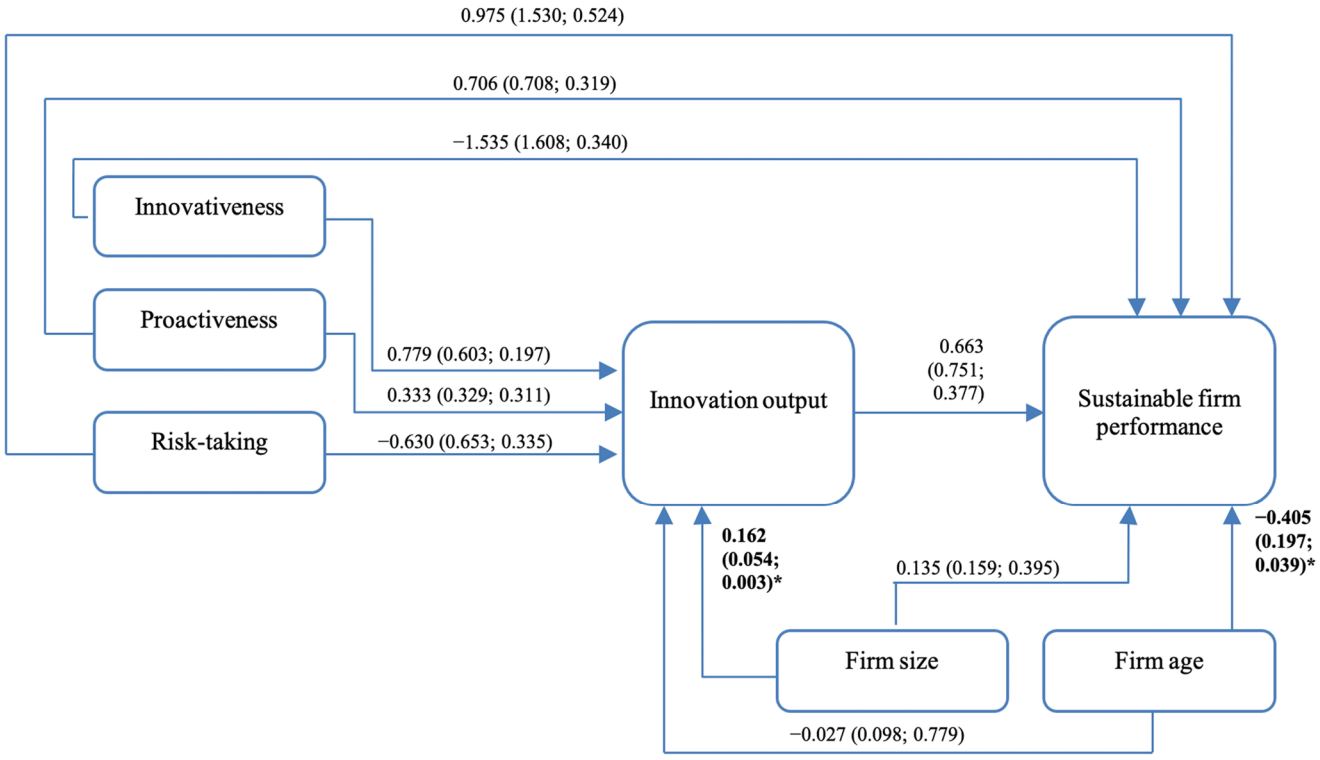



**Figure 3.** Relationships between entrepreneurial orientation, innovation output, and firm performance in non-family firms. Above arrows parameter estimates are given; in brackets standard errors and *p*-values are provided, bold font size and asterisk are used for statistically significant relationships.

All the above suggests a question: are family and non-family businesses significantly different when it comes to relationships between entrepreneurial orientation, innovation output, and sustainable firm performance? The comparison of restricted and unrestricted models shows that the differences, although visible in a detailed analysis of the content of the models, are not significant enough to fully distinguish family and non-family companies, taking into account the context of the study (Poland). The right-tailed probability of the chi-squared distribution was insignificant, with $p = 0.548$ for the model presented in Figures 2 and 3. Thus, from the analytical point of view, constrained and unconstrained models are relatively similar. Although models are not significantly different, it is worthwhile to focus on individual differences between FB and non-FB regarding individual paths and relationships. Focusing on the main implications from the analysis of Figures 2 and 3, we conclude what follows: First, age negatively influences sustainable firm performance in non-family businesses ($\beta = -0.405$; SE = 0.197; $p = 0.039$), while it is unimportant for their family counterparts ($\beta = -0.240$; 0.224; 0.284). Secondly, innovativeness is negatively associated with sustainable firm performance ($\beta = -0.581$; SE = 0.284; $p = 0.041$) if we analyze the whole sample, but it is irrelevant when a sample is divided into two groups of companies. Thirdly, proactiveness is positively associated with sustainable firm performance in family businesses ($\beta = 0.547$; SE = 0.213; $p = 0.010$), while it is unimportant in their non-family counterparts. Fourthly, these study results reveal that risk-taking is irrelevant for sustainable firm performance, both in family and non-family companies, as a part of entrepreneurial orientation. Fifth, the number of innovations is positively related to sustainable firm performance in family businesses ($\beta = 0.431$; SE = 0.178; $p = 0.016$). However, it is not significant in non-family companies ($\beta = 0.663$; SE = 0.751; $p = 0.377$). Finally, innovativeness, as an EO dimension, is positively associated with a number of innovations in family businesses ($\beta = 0.317$; SE= 0.140; $p = 0.024$), but it is not in non-family companies.

## 5. Discussion

The main aim of the research presented in this paper was to determine if the relationships between entrepreneurial orientation, innovation output, and sustainable organizational performance are significantly different in family and non-family businesses. We conducted quantitative empirical research on the sample of 322 Polish organizations, composed of 161 family and non-family business pairs, selected from the database of 10,009 organizations. We used multi-group SEM to analyze relationships between entrepreneurial orientation dimensions, innovation output, and firm performance. Our methodology involved comparing of two models—one created for family and one for non-family businesses. Although estimated models are not significantly different, by focusing on individual differences between family and non-family businesses regarding individual paths and relationships we can indicate the critical role of the proactiveness dimension in shaping firm performance, according to the results of our research. There is no mediating role of innovativeness in the relationship between entrepreneurial orientation and firm performance, neither in the family nor in non-family businesses. A summary of hypothesis testing is provided in Table 6.

**Table 6.** Hypothesis testing results.

| Hypothesis | Is Hypothesis Supported? | Supporting Evidence from the Literature | Supporting Evidence from This Study |
| --- | --- | --- | --- |
| **H1.** *There are no significant differences between the impact of EO on sustainable firm performance in family and non-family businesses.* | YES | Several works point out insignificant differences between family and non-family companies, including [53]. | Right-tailed chi-square difference test is insignificant with $p = 0.548$ |

**Table 6.** *Cont.*

| Hypothesis | Is Hypothesis Supported? | Supporting Evidence from the Literature | Supporting Evidence from This Study |
| --- | --- | --- | --- |
| **H2.** *Innovation output influences sustainable firm performance stronger in family firms, compared to in non-family businesses.* | YES | As Arzubiaga, Maseda, and Iturralde [106] point out, the influence of innovation on sustainable firm performance is stronger in family than in non-family businesses. | Path coefficient in family businesses between innovation output and firm performance is significant (β = 0.431; SE = 0.178; $p$ = 0.016), while it is insignificant in non-family businesses (β = 0.663; SE = 0.751; $p$ = 0.377) |
| **H3.** *EO influences innovation output more intensely in family businesses, compared to in non-family businesses.* | YES | Kammerlander and van Essen [14] claim that family businesses are more innovative than their non-family counterparts. Positive relationship was also found by Price, Stoica, and Bonacella [107]. | Path coefficient in family business between innovativeness (EO dimension) and innovation output is significant (β = 0.317; SE = 0.140; $p$ = 0.024), while in non-family businesses it is insignificant (β = 0.779; SE = 0.603; $p$ = 0.197). |
| **H4.** *Innovation output mediates the relationship between EO and sustainable firm performance in both FB and NFB.* | NO | The proof for the mediating role of innovation in the relationships is shows by Tsou and Chen [108] who point out the mediating effect of innovation in the relationship between digital technology use and firm performance. Additionally, Zehir, Can, and Karaboga [109] confirm mediating effect of innovation on the relationship between EO and firm performance. | Mediating effect of innovation output in the relationship between EO dimension and sustainable firm performance is insignificant. |

Our research results prove there are a few differences between family and non-family businesses concerning the influence of entrepreneurial orientation on innovation and sustainable firm performance. This might be explained considering that family and non-family businesses were established in similar years after the political transformation (all the changes initiated in Poland in the 1990s, which were focused on building a free market, creating a civil society, and democratization). They use the same operation patterns—management methods characteristic for large corporations, which are copied and imitated due to unlimited access to information. This does not exclude the existence of other differences between family and non-family businesses in transition economies. Rebernik and Širec [110], based on a GEM study, found that: "There is a distinct pattern of differences between family businesses and non-family businesses. In all geographic regions, the percentage of highly innovative non-family businesses is higher than that of family businesses. This difference is the biggest in the European economies, where the proportion of highly innovative non-family businesses exceeds the proportion of highly innovative family businesses by 10%" (p. 16). Perhaps the differences between FBs and NFBs are revealed depending on the stage of development. It should be noted that the majority of family businesses are in a pre-succession phase (they were created after the political and economic transformation), and thus have similar roots as their non-family counterparts. We cannot exclude the possibility that differences between companies develop over time, and in developed economies, they might be more evident.

The analysis of the results shows that H1 must be verified positively, as there are no significant differences between the constrained and unconstrained models as measured by the right-tailed probability of the chi-squared distribution. However, getting into the details shows the differential effect of proactivity on sustainable firm performance in family and non-family companies. In the literature, we can find many studies attempting to find the relationship between entrepreneurial orientation and firm performance. Usually, the influence of entrepreneurial orientation on efficiency or growth is determined or moder-

ated by many additional variables. Therefore, it is also worth paying attention to various types of moderators that regulate the relationship between EO and sustainable firm performance [6]. Researchers pay attention mainly to the influence of the environment [111]. Still, it is possible to indicate other potential moderators, such as organizational culture, organizational climate, transformational leadership [112], managerial power [113], and organization design, especially in the dimensions of the degree of centralization and the level of formalization or organizational strategy.

Referring to the second hypothesis (H2), our research results reveal significant differences between family and non-family businesses regarding the influence of innovation output on sustainable firm performance. These two constructs were significant in family businesses and insignificant in non-family ones. Thus, it brings support to our hypothesis H2. However, the relationship is diverse from what we initially supposed. The average levels of innovation and performance in family and non-family businesses are very similar. Thus, the difference has to be attributed to the diverse influence of innovation on performance in these two types of companies. This clearly shows the necessity of in-depth qualitative-based studies in this respect. Our results prove that innovation is a significantly more robust predictor of performance in family businesses than in non-family ones.

Concerning hypothesis H3, innovativeness as a dimension of entrepreneurial orientation influences the number of innovations in family businesses, but the number of new products or solutions is nonsignificant in non-family businesses. Both innovativeness and risk-taking dimensions play an unimportant role in shaping the number of new products, services, or organizational solutions. It signifies that orientation on proactiveness in family businesses leads to more innovations than in non-family businesses. Proactiveness helps implement novel and useful ideas, proven by our results. Therefore, managers and owners of a family business should focus their attention on strengthening proactiveness by formulating goals that may appear bold and challenging—such as introducing new products or services, new ventures, and entering new markets; however, proactiveness strategies seem to lead to innovative actions and outcomes [114]. It is worth emphasizing that proactiveness does not just refer to responsive behavior against the market environment, but is related to careful observation of the context of the opportunities it offers and the ability to anticipate future needs and expectations [115]. The organization should learn to act in advance, not only to react to changes or phenomena that have already occurred, which leads to being "first in the market" [98].

Our research results show no mediating role of some innovations in the relationship between entrepreneurial orientation and sustainable firm performance (H4), neither in the family nor in non-family businesses. This falsifies assumptions indicated in hypothesis H4. This might be due to the distant relationship between innovation and sustainable firm performance, as measured in our research [116]. This research shows a positive correlation between entrepreneurial orientation and firm performance. However, it is doubtful whether this dependence is constant over time or whether the positive result of the activity is only temporary. Wiklund [96] shows that this relationship strengthens over time, and investments in entrepreneurial orientation pay off in the future. A critical issue indicated by some researchers [117] is how to measure innovation output, as many indicators seem to show some weaknesses, which can cause research bias. Contradictory to what is suggested in Oslo Manual [56], we suggest using relative indicators to measure innovation output. The ratio of the number of innovations to the number of employees would be a better indicator of the innovativeness of a company than the number of innovations per se. Further innovation output research also seems necessary to consider the types of innovation—radical and incremental.

The lack of significant differences between models might be explained by studying the very phenomenon of entrepreneurial orientation. It exists in both family and non-family firms [118], thus its nature remains to some extent the same. Although, the impact of different dimensions of entrepreneurial orientation on innovation and sustainable firm performance might be slightly different, and that is what we observed in our study. Thus,

we dug into the data in searching for small, albeit important, differences, and the one we were able to locate is related to the influence of proactivity on sustainable firm performance, which was significant for family firms and insignificant for non-family firms. The same applies to the path between innovation and sustainable firm performance, which was important in family and unimportant in non-family firms. However, these dissimilarities do not alter models enough to claim that the two created models are significantly different.

When interpreting the results of our research, it should also be emphasized that family firms are not a homogeneous set, and their heterogeneity has not only been confirmed in numerous studies [119,120], but research also shows implications for the management of FBs [121]. According to our study findings, evidently, in most cases, larger companies generate more innovations. This might hinder the research results while considering the size as a control variable. The research requires a more detailed examination of company size as a variable for the EO–performance relationship [1,6,105,122].

Like other researchers [123], we believe that contextual variables, including those describing the context of a given country or region, deserve special attention in further research. Individual countries are characterized by specific cultural, political, or institutional differences that affect entrepreneurial behavior [123]. These differences may additionally result from the current stage of socio-economic development, the transformation of the economy, and the accompanying social changes taking place in countries such as Poland. Compared to the transforming economies (CEE countries), mature economies create different conditions for entrepreneurs to operate; the level of development of innovative ecosystems is different, and the level of trust as social capital or the level of entrepreneurial intentions is also different.

## 6. Conclusions

### *6.1. Summary*

The main purpose of the research presented in this article was to examine the relationship between entrepreneurial orientation and firm performance, taking into account the mediating role of innovation in family and non-family enterprises in the post-transformation economy. Quantitative research on a sample of 322 Polish organizations was carried out to reach that aim. The research results indicate few differences between family and non-family enterprises in the case of the impact of entrepreneurial orientation on innovativeness and firm performance. The study's outcomes contribute to academic research and business practice. First, the study's results support the theory of FBs, as they reveal the differences between FBs and NFBs concerning the effect of proactivity on firm performance. Second, by showing that innovativeness, perceived as a posture in FBs, helps implement novel and valuable ideas. The paper provides arguments for FBs' and NFBs' managers to focus on strengthening firms' proactiveness.

### *6.2. Theoretical Implications*

Differences between FBs and NFBs, or the lack of them, may result from family businesses' characteristics in a given country. In Poland, there are no differences between family and non-family firms in terms of the impact of EO on firms' profitability, as there are many similarities between those two groups of companies. It is mainly because FBs and NFBs were established after the transformation in 1989. Thus, many family firms in the Polish market have not yet passed their first succession. So, these businesses are comparable in many ways, e.g., our previous research did not show any differences in the level of their innovativeness [124]. The presented research shows how complex the relationships between EO and sustainable firm performance are, especially when considered in the context of implemented innovations. This indicates the need for further research on these relationships. Our research results prove that the orientation of a company, no matter the strength and direction, has a limited effect on actual performance measures, such as innovation output—i.e., the number of novel and valuable products, services, management methods, or sustainable firm performance. This is in line with the arguments

and research results of Rezaei and Ortt [125]. They point to the fact that although EO is usually positively related to performance, such a relationship should be explained more complexly, distinguishing different types of performance connected strictly to different functions of a company, as EO dimensions are related to them in diverse ways. Managers striving to increase a firm's sustainability should focus not only on overall performance, but also on types of functional firm performances. What is more, it is worth emphasizing that focusing on EO cannot only be declarative, but requires the management to commit and take actual actions to increase performance [126].

Trying to explain the relatively weak relationship between EO dimensions and innovation output is necessary to consider the role of financing novel and valuable organizational solutions. Focusing on entrepreneurial posture and mindset, as represented by the EO dimensions, does not guarantee the introduction of new products, services, or practices. For most innovations in organizations, financial resources or, in more extreme cases, availability and access to external financing are critical. To a large extent, previous profits and good relationships with investors seem to affect the effectiveness of innovation activities. As a result, the relationship between innovation and performance may be treated as mutual, which means that these organizational outcomes are, in fact, interrelated. On the other hand, this does not exclude the possibility of testing a reversed relationship, where sustainable performance and the ability to generate profits influence innovation output [127].

It is also worth noting that key decision makers' actual postures and behaviors, especially in small- and medium-sized organizations, are frequently affected by numerous cognitive biases [128]. Researchers must include the influence of individual differences in management styles, educational background, and previous experience of key players in smaller companies in explaining the managerial approach to innovation [129]. This influence was not tested in our research and requires empirical evidence.

From the theoretical point of view, these research results bring insight into the role of the organizational context and firm characteristics. The lack of significant differences between the model created for family and non-family businesses might be explained by considering their similarities in size and, to a large extent, profile. With the increased access to information and large transparency of organizational actions, it is relatively easy for companies to compete with market rivals successfully and imitate effective actions of larger or more professionally managed companies [130]. This leads to a larger homogeneity of organizations in terms of modes of operations. Previously reserved for the relatively small group of well-managed companies, organizational practices and policies are now available to every decision maker. In addition, the transition of effective practices has occurred within a certain industry and is now available for companies operating in different contexts. Through benchmarking and imitation, companies became more similar, and differences in their effectiveness when implementing innovations may lie in discrete, unobservable behaviors, employees' competencies, or abilities to make informed decisions.

### *6.3. Practical Implications*

Thus, companies, no matter the size and age, should focus more on implementing novel and useful ideas, especially considering the effect of innovation output on firm performance in family businesses. The situation is more complex in non-family businesses—we did not observe a direct effect of innovation on performance in such companies. This might be due to the temporal separation of cause and effect by which we perceive innovation and firm performance [131]. As Shouyu [132] indicates, the relationship between these two constructs may be indirect, as companies achieve better performance through innovation in diverse ways.

Our study suggests that an especially pro-innovative posture manifested in entrepreneurial orientation in family businesses leads to an increased number of novel and valuable solutions, and new products and services. Thus, in family businesses, creating the proper atmosphere and mindset for innovations should result in more innovations. On the other hand, research results also prove the link between proactiveness and sustainable firm

performance. This reaffirms that commitment and personal characteristics are important in driving organizational results, especially in SMEs [45].

The negative influence of firm age on non-family sustainable firm performance draws attention to the problem of older, already experienced, and market-valued companies being less productive. This might be related to the fact that more experienced companies tend to perceive their financial and market standings with more reserve, but it also might mean that younger companies have what it takes to increase their market share. This is most likely due to their higher capabilities to change under adversities and the higher commitment of managers to the everyday life of smaller and younger companies. On the other hand, larger companies report a higher number of innovations (which is easily explainable considering the sheer size, number of employees and ideas generated within, and their larger financial potential to introduce changes). Interestingly, these relationships are insignificant in the family business. This might be due to the commitment of owners of family SMEs to the functioning of organizations, but also their particular and frequently loudly expressed interest in the well-being of a family firm [40]. Ownership, control, and responsibility for family explain the lack of influence of organizational ageing on the performance and innovativeness of family businesses. These postures shall be imitated by non-family businesses to mitigate the negative impact of a firm's age on performance.

### *6.4. Study Limitations*

It should be noted that the identification of relationships between innovation and the achievements of enterprises is related to the methods used to measure them, and the results of the research may be affected by the measures (measures/indicators) of both innovation and the results of enterprises (performance measurement).

When considering the results of the study presented in the article, it should be taken into account that it was conducted during a pandemic. In this period, many companies were forced to make extraordinary decisions, often deviating from the ones they usually make. Hence, the study results would have been different in other market conditions.

Although the results of the presented research shed some light on the role of entrepreneurial orientation in shaping sustainable performance, the research sample was selected from a set of family and non-family enterprises operating in Poland. Therefore, potential generalizations of the obtained results should be made with caution.

### *6.5. Suggestions for Future Work*

In further research on the relationship between EO and sustainable firm performance at FBs and NFBs, it is worth considering such moderating variables as the level of public support for entrepreneurs, the efficiency of public administration, or other (e.g., national culture). Our research does not provide an unambiguous answer as to whether and to what extent the specificity of the economic development stage influenced their results. Transformation processes and their effectiveness from the point of view of the entrepreneurial orientation of companies play an important role. Therefore, further research is necessary.

Various firm performance indicators are closely related to the multidimensionality of this construct [133]. The relationship between entrepreneurial orientation and firm performance may depend on the indicators used to evaluate them. Taking into account the conceptualizations most often offered in previous studies, the measures of performance could be supplemented with non-financial ones [134], such as enterprise development and profitability [135], or improvement in customer satisfaction, employees' satisfaction, product quality, and service quality [136]. Moreover, many factors besides entrepreneurial orientation affect the firm performance, and it is difficult to isolate their influence [6].

The cross-sectional character of conducted research has limited our ability to reliably infer the relationship between entrepreneurial orientation and sustainable firm performance. Providing stronger arguments regarding the cause-and-effect relationship between entrepreneurial orientation and firm performance, especially the occurring delays, requires a long-term series of measurements of both variables. However, implementing such a

research intention is a real challenge both because of the difficulties in obtaining information in repetitive periods and because of changes in the studied population due to the mergers, acquisitions, and bankruptcy of enterprises. This is particularly important because the cause-and-effect relationship presented in this paper can be reversed. Lumpkin [137], reflecting on increasing the influence of entrepreneurship as a research domain, proposes moving away from dependent variables related to financial efficiency indicators and paying attention to entrepreneurship's role in learning, adaptation, change, and creativity. This hint suggests reversing the direction of the relationship between entrepreneurial orientation and performance: better performance might also stimulate entrepreneurial orientation [6].

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## References

1. Lumpkin, G.T.; Dess, G.G. Clarifying the entrepreneurial orientation construct and linking it to performance. *Acad. Manag. Rev.* **1996**, *21*, 135–172. [CrossRef]
2. Anderson, B.S.; Kreiser, P.M.; Kuratko, D.F.; Hornsby, J.S.; Eshima, Y. Reconceptualizing entrepreneurial orientation. *Strateg. Manag. J.* **2015**, *36*, 1579–1596. [CrossRef]
3. Covin, J.G.; Wales, W.J. Crafting high-impact entrepreneurial orientation research: Some suggested guidelines. *Entrep. Theory Pract.* **2019**, *43*, 3–18. [CrossRef]
4. Miller, D. The correlates of entrepreneurship in three types of firms. *Manag. Sci.* **1983**, *29*, 770–791. [CrossRef]
5. Pimentel, D.; Couto, J.P.; Scholten, M. Entrepreneurial orientation in family firms: Looking at a European outermost region. *J. Enter. Cult.* **2017**, *25*, 441–460. [CrossRef]
6. Rauch, A.; Wiklund, J.; Lumpkin, G.T.; Frese, M. Entrepreneurial orientation and business performance: An assessment of past research and suggestions for the future. *Entrep. Theory Pract.* **2009**, *33*, 761–787. [CrossRef]
7. Wales, W.J.; Gupta, V.K.; Marino, L.; Shirokova, G. Entrepreneurial orientation: International, global and cross-cultural research. *Int. Small Bus. J.* **2019**, *37*, 95–104. [CrossRef]
8. Wiklund, J.; Shepherd, D. Knowledge-based resources, entrepreneurial orientation, and the performance of small and medium-sized businesses. *Strateg. Manag. J.* **2003**, *24*, 1307–1314. [CrossRef]
9. Covin, J.G.; Slevin, D.P. Strategic management of small firms in hostile and benign environments. *Strateg. Manag. J.* **1989**, *10*, 75–87. [CrossRef]
10. Covin, J.G.; Green, K.M.; Slevin, D.P. Strategic process effects on the entrepreneurial orientation-sales growth rate relationship. *Entrep. Theory Pract.* **2006**, *30*, 57–81. [CrossRef]
11. Miller, D.; Le Breton-Miller, I. Governance, social identity, and entrepreneurial orientation in closely held public companies. *Entrep. Theory Pract.* **2011**, *35*, 1051–1076. [CrossRef]
12. Hernández-Linares, R.; López-Fernández, M.C. Entrepreneurial orientation and the family firm: Mapping the field and tracing a path for future research. *Fam. Bus. Rev.* **2018**, *31*, 318–351. [CrossRef]

13. Arzubiaga, U.; Kotlar, J.; De Massis, A.; Maseda, A.; Iturralde, T. Entrepreneurial orientation and innovation in family SMEs: Unveiling the (actual) impact of the Board of Directors. *J. Bus. Vent.* **2018**, *33*, 455–469. [CrossRef]
14. Kammerlander, N.; van Essen, M. Family firms are more innovative than other companies. *Harv. Bus. Rev.* **2017**, *25*, 1–5.
15. Wang, E.S.T.; Juan, P.Y. Entrepreneurial orientation and service innovation on consumer response: A B & B Case. *J. Small Bus. Manag.* **2016**, *54*, 532–545.
16. Titus, V.; Parker, O.; Covin, J. Organizational aspirations and external venturing: The contingency of entrepreneurial orientation. *Entrep. Theory Pract.* **2020**, *44*, 645–670. [CrossRef]
17. Yu, W.; Wiklund, J.; Pérez-Luño, A. ADHD symptoms, entrepreneurial orientation (EO) and firm performance. *Entrep. Theory Pract.* **2021**, *45*, 92–117. [CrossRef]
18. Eliakis, S.; Kotsopoulos, D.; Karagiannaki, A.; Pramatari, K. Survival and Growth in Innovative Technology Entrepreneurship: A Mixed-Methods Investigation. *Adm. Sci.* **2020**, *10*, 39. [CrossRef]
19. Gunday, G.; Ulusoy, G.; Kilic, K.; Alpkan, L. Effects of innovation types on firm performance. *Int. J. Prod. Econ.* **2011**, *133*, 662–676. [CrossRef]
20. Kemp, R.G.; Folkeringa, M.; De Jong, J.P.; Wubben, E.F. Research Report H200207. SCALES SCientific Analysis of Entrepreneurship and SMEs, The Netherlands. In *Innovation and Firm Performance*; EIM Business and Policy Research Location: Rotterdam, The Netherlands, 2003; p. 68. Available online: www.eim.nl/smes-and-entrepreneurship (accessed on 1 December 2022).
21. Kotsopoulos, D.; Karagianaki, A.; Baloutsos, S. The effect of human capital, innovation capacity, and COVID-19 crisis on Knowledge-Intensive Enterprises' growth within a VC-driven innovation ecosystem. *J. Bus. Res.* **2022**, *139*, 1177–1191. [CrossRef]
22. Bruton, G.D.; Zahra, A.A.; Cai, L. Examining entrepreneurship through indigenous lenses. *Entrep. Theory Pract.* **2018**, *42*, 351–361. [CrossRef]
23. Martens, C.D.P.; Lacerda, F.M.; Belfort, A.C.; de Freitas, H.M.R. Research on entrepreneurial orientation: Current status and future agenda. *Int. J. Entrep. Behav. Res.* **2016**, *22*, 1–28. [CrossRef]
24. Polska Agencja Rozwoju Przedsiębiorczości/Polish Agency for Enterprise Development. *Family Businesses and the Polish Economy—Opportunities and Challenges*; Research Report; Polska Agencja Rozwoju Przedsiębiorczości: Warsaw, Poland, 2009; p. 189. Available online: https://www.parp.gov.pl/storage/publications/pdf/2011_sme_report_pl.pdf (accessed on 1 December 2022).
25. Rothausen, T.J. "Family" in organizational research: A review and comparison of definitions and measures. *J. Organ. Behav.* **1999**, *20*, 817–836. [CrossRef]
26. Jaskiewicz, P.; Combs, J.G.; Shanine, K.K.; Kacmar, K.M. Introducing the family: A review of family science with implications for management research. *Acad. Manag. Ann.* **2017**, *11*, 309–341. [CrossRef]
27. Hernández-Perlines, F.H.; Cisneros, M.A.I. The role of environment in sustainable entrepreneurial orientation. The case of family firms sustainability. *Sustainability* **2018**, *10*, 2037. [CrossRef]
28. Daspit, J.J.; Chrisman, J.J.; Sharma, P.; Pearson, A.W.; Long, R.G. A strategic management perspective of the family firm: Past trends, new insights, and future directions. *J. Manag. Issues* **2017**, *29*, 6–29.
29. Mączyńska, M. *Firmy Rodzinne. Nowa Odkryta Siła Polskiej Gospodarki. Raport Sektora o Stanie MSP w Polsce [Family Businesses. The New Discovered Strength of the Polish Economy. Sector Report on the Condition of SMEs in Poland]*; PARP: Warszawa, Poland, 2016. Available online: https://www.parp.gov.pl/storage/publications/pdf/2016_raport_msp_pl_clik.pdf (accessed on 1 December 2022).
30. Chrisman, J.J.; Chua, J.H.; Kellermanns, F.W. Priorities, resource stocks, and performance in family and non-family firms. *Entrep. Theory Pract.* **2009**, *33*, 739–760. [CrossRef]
31. Kosmidou, V.; Ahuja, M.K. A configurational approach to family firm innovation. *Fam. Bus. Rev.* **2019**, *32*, 154–173. [CrossRef]
32. De Massis, A.; Di Minin, A.; Frattini, F. Family-driven innovation: Resolving the paradox in family firms. *Calif. Manag. Rev.* **2015**, *58*, 5–19. [CrossRef]
33. Alberti, F.G.; Pizzurno, E. Technology, innovation and performance in family firms. *Int. J. Entrep. Innov. Manag.* **2013**, *17*, 142–161. [CrossRef]
34. Cesaroni, F.M.; Chamochumbi, G.D.; Sentuti, A. Family firms and innovation from founder to successor. *Adm. Sci.* **2021**, *11*, 54. [CrossRef]
35. Werner, A.; Schröder, C.; Chlosta, S. Driving factors of innovation in family and non-family SMEs. *Small Bus. Econ.* **2018**, *50*, 201–218. [CrossRef]
36. Basco, R.; Hernández-Perlines, F.; Rodríguez-García, M. The effect of entrepreneurial orientation on firm performance: A multigroup analysis comparing China, Mexico, and Spain. *J. Bus. Res.* **2020**, *113*, 409–421. [CrossRef]
37. Habib, M.A.; Bao, Y.; Nabi, N.; Dulal, M.; Asha, A.A.; Islam, M. Impact of strategic orientations on the implementation of green supply chain management practices and sustainable firm performance. *Sustainability* **2021**, *13*, 340. [CrossRef]
38. Steinerowska-Streb, I.; Kraśnicka, T. Orientacja przedsiębiorcza a wyniki polskich przedsiębiorstw [Entrepreneurial Orientation and Business Performance of Polish Companies]. *Prz. Organ. Organ. Rev.* **2022**, *1*, 19–29.
39. Lee, T.; Chu, W. The relationship between entrepreneurial orientation and firm performance: Influence of family governance. *J. Fam. Bus. Strat.* **2017**, *8*, 213–223. [CrossRef]
40. Hernandez-Perlines, F.H.; Covin, J.G.; Ribeiro-Soriano, D.E. Entrepreneurial orientation, concern for socioemotional wealth preservation, and family firm performance. *J. Bus. Res.* **2021**, *126*, 197–208. [CrossRef]

41. Arzubiaga, U.; Iturralde, T.; Maseda, A.; Kotlar, J. Entrepreneurial orientation and firm performance in family SMEs: The moderating effects of family, women, and strategic involvement in the board of directors. *Int. Entrep. Manag. J.* **2018**, *14*, 217–244. [CrossRef]
42. Calabrò, A.; Vecchiarini, M.; Gast, J.; Campopiano, G.; De Massis, A.; Kraus, S. Innovation in family firms: A systematic literature review and guidance for future research. *Int. J. Manag. Rev.* **2019**, *21*, 317–355. [CrossRef]
43. Lumpkin, G.T.; Brigham, K.H.; Moss, T.W. Long-term orientation: Implications for the entrepreneurial orientation and performance of family businesses. *Entrep. Reg. Dev.* **2010**, *22*, 241–264. [CrossRef]
44. Alayo, M.; Maseda, A.; Iturralde, T.; Arzubiaga, U. Internationalization and entrepreneurial orientation of family SMEs: The influence of the family character. *Int. Bus. Rev.* **2019**, *28*, 48–59. [CrossRef]
45. Hernández-Linares, R.; Kellermanns, F.W.; López-Fernández, M.C.; Sarkar, S. The effect of socioemotional wealth on the relationship between entrepreneurial orientation and family business performance. *BRQ Bus. Res. Q.* **2020**, *23*, 174–192. [CrossRef]
46. Steinerowska-Streb, I.; Wziątek-Staśko, A. Innovations in family firms: A study of owner-managers' knowledge development. *J. Fam. Bus. Manag.* **2020**, *10*, 247–264. [CrossRef]
47. Beck, L.; Janssens, W.; Debruyne, M.; Lommelen, T. A study of the relationships between generation, market orientation, and innovation in family firms. *Fam. Bus. Rev.* **2011**, *24*, 252–272. [CrossRef]
48. Kellermanns, F.W.; Eddleston, K.A.; Barnett, T.; Pearson, A. An exploratory study of family member characteristics and involvement: Effects on entrepreneurial behavior in the family firm. *Fam. Bus. Rev.* **2008**, *21*, 1–14. [CrossRef]
49. Classen, N.; Carree, M.; Van Gils, A.; Peters, B. Innovation in family and non-family SMEs: An exploratory analysis. *Small Bus. Econ.* **2014**, *42*, 595–609. [CrossRef]
50. Pittino, D.; Martínez, A.B.; Chirico, F.; Galván, R.S. Psychological ownership, knowledge sharing and entrepreneurial orientation in family firms: The moderating role of governance heterogeneity. *J. Bus. Res.* **2018**, *84*, 312–326. [CrossRef]
51. Serrano-Bedia, A.M.; López-Fernández, M.C.; Garcia-Piqueres, G. Analysis of the relationship between sources of knowledge and innovation performance in family firms. *Innovation* **2016**, *18*, 489–512. [CrossRef]
52. Hiebl, M.R. Risk aversion in the family business: The dark side of caution. *J. Bus. Strategy* **2014**, *35*, 38–42. [CrossRef]
53. Coleman, S.; Cotei, C.; Farhat, J. A resource-based view of new firm survival: New perspectives on the role of industry and exit route. *J. Dev. Entrep.* **2013**, *18*, 1350002. [CrossRef]
54. Manzaneque, M.; Ramírez, Y.; Diéguez-Soto, J. Intellectual capital efficiency, technological innovation and family management. *Innovation* **2017**, *19*, 167–188. [CrossRef]
55. Tajeddini, K.; Ratten, V. The moderating effect of brand orientation on inter-firm market orientation and performance. *J. Strat. Mark.* **2020**, *28*, 194–224. [CrossRef]
56. OECD. Oslo Manual. In *The Measurement of Scientific, Technological and Innovation Activities: Guidelines for Collecting, Reporting and Using Data on Innovation*; OECD: Paris, France, 2018.
57. Lopez-Fernandez, M.C.; Serrano-Bedia, A.M.; Gómez-López, R. Determinants of innovation decision in small and medium-sized family enterprises. *J. Small Bus. Ent. Dev.* **2016**, *23*, 408–427. [CrossRef]
58. Duran, P.; Kammerlander, N.; van Essen, M.; Zellweger, T. Doing more with less: Innovation input and output in family firms. *Acad. Manag. J.* **2016**, *59*, 1224–1264. [CrossRef]
59. Kellermanns, F.W.; Eddleston, K.A.; Sarathy, R.; Murphy, F. Innovativeness in family firms: A family influence perspective. *Small Bus. Econ.* **2012**, *38*, 85–101. [CrossRef]
60. Diéguez-Soto, J.; Manzaneque, M.; Rojo-Ramirez, A.A. Technological innovation inputs, outputs, and performance: The moderating role of family involvement in management. *Fam. Bus. Rev.* **2016**, *29*, 327–346. [CrossRef]
61. Kammerlander, N.; Patzelt, H.; Behrens, J.; Röhm, C. Organizational ambidexterity in family-managed firms: The role of family involvement in top management. *Fam. Bus. Rev.* **2020**, *33*, 393–423. [CrossRef]
62. Kraśnicka, T.; Steinerowska-Streb, I. Family involvement and innovation in family enterprises. Zeszyty Naukowe Politechniki Śląskiej. Organizacja i Zarządzanie. *Sci. J. Sil. Univ. Technol. Organ. Manag.* **2019**, *136*, 289–303.
63. Block, J.H. R & D investments in family and founder firms: An agency perspective. *J. Bus. Vent.* **2012**, *27*, 248–265.
64. Llach, J.; Nordqvist, M. Innovation in family and non-family businesses: A resource perspective. *Int. J. Entrep. Ventur.* **2010**, *2*, 381–399. [CrossRef]
65. Muñoz-Bullón, F.; Sanchez-Bueno, M.J.; De Massis, A. Combining internal and external R & D: The effects on innovation performance in family and nonfamily firms. *Entrep. Theory Pract.* **2020**, *44*, 996–1031.
66. Becerra, M.; Cruz, C.; Graves, C. Innovation in family firms: The relative effects of wealth concentration versus family-centered goals. *Fam. Bus. Rev.* **2020**, *33*, 372–392. [CrossRef]
67. Brinkerink, J. Broad search, deep search, and the absorptive capacity performance of family and nonfamily firm R & D. *Fam. Bus. Rev.* **2018**, *31*, 295–317.
68. Hyytinen, A.; Pajarinen, M.; Rouvinen, P. Does innovativeness reduce startup survival rates? *J. Bus. Ventur.* **2015**, *30*, 564–581. [CrossRef]
69. Rosli, M.M.; Sidek, S. The Impact of innovation on the performance of small and medium manufacturing enterprises: Evidence from Malaysia. *J. Innov. Manag. Small Medium Enterp.* **2013**, *1*, 885666. [CrossRef]

70. Rosenbusch, N.; Brinckmann, J.; Bausch, A. Is innovation always beneficial? A meta-analysis of the relationship between innovation and performance in SMEs. *J. Bus. Ventur.* **2011**, *26*, 441–457. [CrossRef]
71. Fuetsch, E.; Suess-Reyes, J. Research on innovation in family businesses: Are we building an ivory tower? *J. Fam. Bus. Manag.* **2017**, *7*, 44–92. [CrossRef]
72. Jaskiewicz, P.; Combs, J.G.; Rau, S.B. Entrepreneurial legacy: Toward a theory of how some family firms nurture transgenerational entrepreneurship. *J. Bus. Ventur.* **2015**, *30*, 29–49. [CrossRef]
73. Steeger, J.H.; Hoffmann, M. Innovation and family firms: Ability and willingness and German SMEs. *J. Fam. Bus. Manag.* **2016**, *6*, 251–269. [CrossRef]
74. Chrisman, J.J.; Chua, J.H.; De Massis, A.; Frattini, F.; Wright, M. The ability and willingness paradox in family firm innovation. *J. Prod. Innov. Manag.* **2015**, *32*, 310–318. [CrossRef]
75. Astrachan, J.H. Strategy in family business: Toward a multidimensional research agenda. *J. Fam. Bus. Strategy* **2010**, *1*, 6–14. [CrossRef]
76. Pérez-Luño, A.; Wiklund, J.; Cabrera, R.V. The dual nature of innovative activity: How entrepreneurial orientation influences innovation generation and adoption. *J. Bus. Ventur.* **2011**, *26*, 555–571. [CrossRef]
77. Meroño-Cerdán, A.L.; López-Nicolás, C.; Molina-Castillo, F.J. Risk aversion, innovation and performance in family firms. *Econ. Innov. New Tech.* **2018**, *27*, 189–203. [CrossRef]
78. Alegre, J.; Chiva, R. Linking entrepreneurial orientation and firm performance: The role of organizational learning capability and innovation performance. *J. Small Bus. Manag.* **2013**, *51*, 491–507. [CrossRef]
79. Sanchez-Famoso, V.; Maseda, A.; Iturralde, T. Family involvement in top management team: Impact on relationships between internal social capital and innovation. *J. Manag. Organ.* **2017**, *23*, 136–162. [CrossRef]
80. Wales, W.J.; Gupta, V.K.; Mousa, F.T. Empirical research on entrepreneurial orientation: An assessment and suggestions for future research. *Int. Small Bus. J.* **2013**, *31*, 357–383. [CrossRef]
81. Roxas, B.; Ashill, N.; Chadee, D. Effects of entrepreneurial and environmental sustainability orientations on firm performance: A study of small businesses in the Philippines. *J. Small Bus. Manag.* **2017**, *55*, 163–178. [CrossRef]
82. Eisenhardt, K.M.; Santos, F.M. Knowledge-based view: A new theory of strategy. *Handb. Strategy Manag.* **2002**, *1*, 139–164.
83. Wang, C.L. Entrepreneurial orientation, learning orientation, and firm performance. *Entrep. Theory Pract.* **2008**, *32*, 635–657. [CrossRef]
84. Asad, M.; Shabbir, M.; Salman, R.; Haider, S.; Ahmad, I. Do entrepreneurial orientation and size of enterprise influence the performance of micro and small enterprises? A study on mediating role of innovation. *Manag. Sci. Let.* **2018**, *8*, 1015–1026. [CrossRef]
85. Soares, M.D.C.; Perin, M.G. Entrepreneurial orientation and firm performance: An updated meta-analysis. *RAUSP Manag. J.* **2020**, *55*, 143–159. [CrossRef]
86. Hilami, M.F.; Ramayah, T.; Mustapha, Y.; Pawanchik, S. Product and process innovativeness: Evidence from Malaysian SMEs. *Eur. J. Soc. Sci.* **2010**, *16*, 557–568.
87. Amann, B.; Jaussaud, J. Family and non-family business resilience in an economic downturn. *Asia Pac. Bus. Rev.* **2012**, *18*, 203–223. [CrossRef]
88. Surdej, A. *National Report on Family Businesses in Poland*; Cracow University of Economics: Cracow, Poland, 2015; Available online: https://www.insist-project.eu/index.php/files/400/Cases-Reports-and-Study-by-the-INSIST-project/28/PL-National-Report-on-Family-Businesses---Poland.pdf (accessed on 1 December 2022).
89. Cronbach, L.J. Coefficient alpha and the internal structure of tests. *Psychometrika* **1951**, *16*, 297–334. [CrossRef]
90. Antoncic, B.; Hisrich, R.D. Intrapreneurship: Construct refinement and cross-cultural validation. *J. Bus. Ventur.* **2001**, *16*, 495–527. [CrossRef]
91. Kreiser, P.M.; Marino, L.D.; Weaver, K.M. Assessing the psychometric properties of the entrepreneurial orientation scale: A multi-country analysis. *Entrep. Theory Pract.* **2002**, *26*, 71–93. [CrossRef]
92. Wójcik-Karpacz, A.; Kraus, S.; Karpacz, J. Examining the relationship between team-level entrepreneurial orientation and team performance. *Int. J. Entrep. Behav. Res.* **2022**, *28*, 1–30. [CrossRef]
93. Kalnins, A. Multicollinearity: How common factors cause Type 1 errors in multivariate regression. *Strateg. Manag. J.* **2018**, *39*, 2362–2385. [CrossRef]
94. Muthen, L.K.; Muthen, B.O. *Mplus Statisical Analysis with Latent Variables: User's Guide*, 7th ed.; Muthen & Muthen: Los Angeles, CA, USA, 2012.
95. Yuan, K.H.; Chan, W. Measurement invariance via multigroup SEM: Issues and solutions with chi-square-difference tests. *Psychol. Methods* **2016**, *21*, 405. [CrossRef]
96. Wiklund, J. The sustainability of the entrepreneurial orientation-performance relationship. *Entrep. Theory Pract.* **1999**, *24*, 37–48. [CrossRef]
97. Dess, G.G.; Lumpkin, G.T. The role of entrepreneurial orientation in stimulating effective corporate entrepreneurship. *Acad. Manag. Exec.* **2005**, *19*, 147–156. [CrossRef]
98. Lumpkin, G.T.; Dess, G.G. Linking two dimensions of entrepreneurial orientation to firm performance: The moderating role of environment and industry life cycle. *J. Bus. Ventur.* **2001**, *16*, 429–451. [CrossRef]

99. Covin, J.G.; Slevin, D.P. The influence of organization structure on the utility of an entrepreneurial top management style. *Strateg. Manag. J.* **1988**, *25*, 217–234. [CrossRef]
100. Bratnicki, M.; Gabryś, B.J. Orientacja przedsiębiorcza a sukces organizacji [Entrepreneurial Orientation and Organization's Success]. In *Prace i Materiały Wydziału Zarządzania Uniwersytetu Gdańskiego [Works Materials of the Faculty of Management of the University of Gdańsk]*; Faculty of Management of the University of Gdansk: Gdansk, Poland, 2011; pp. 63–72.
101. Dyduch, W. *Pomiar Przedsiębiorczości Organizacyjnej [Entrepreneurial Orientation Measurement]*; Wydawnictwo Akademii Ekonomicznej: Katowice, Poland, 2008.
102. Damanpour, F. Footnotes to research on management innovation. *Organ. Stud.* **2014**, *35*, 1265–1285. [CrossRef]
103. Su, Z.; Xie, E.; Li, Y. Entrepreneurial orientation and firm performance in new ventures and established firms. *J. Small Bus. Manag.* **2011**, *49*, 558–577. [CrossRef]
104. Herrera, L.; Sánchez-González, G. Firm size and innovation policy. *Int. Small Bus. J.* **2013**, *31*, 137–155. [CrossRef]
105. Wales, W.J.; Wiklund, J.; McKelvie, A. What about new entry? Examining the theorized role of new entry in the entrepreneurial orientation–performance relationship. *Int. Small Bus. J.* **2015**, *33*, 351–373. [CrossRef]
106. Arzubiaga, U.; Maseda, A.; Iturralde, T. Exploratory and exploitative innovation in family businesses: The moderating role of the family firm image and family involvement in top management. *Rev. Manag. Sci.* **2019**, *13*, 1–31. [CrossRef]
107. Price, D.P.; Stoica, M.; Boncella, R.J. The relationship between innovation, knowledge, and performance in family and non-family firms: An analysis of SMEs. *J. Innov. Entrep.* **2013**, *2*, 1–20. [CrossRef]
108. Tsou, H.T.; Chen, J.S. How does digital technology usage benefit firm performance? Digital transformation strategy and organisational innovation as mediators. *Technol. Anal. Strateg. Manag.* **2021**, 1–14. [CrossRef]
109. Zehir, C.; Can, E.; Karaboga, T. Linking entrepreneurial orientation to firm performance: The role of differentiation strategy and innovation performance. *Procedia Soc. Behav. Sci.* **2015**, *210*, 358–367. [CrossRef]
110. Rebernik, M.; Širec, K. *Resilience of Entrepreneurial Activity: GEM Slovenia 2021, Executive Summary*; University of Maribor, University Press: Maribor, Slovenia, 2022.
111. Covin, J.G.; Lumpkin, G.T. Entrepreneurial orientation theory and research: Reflections on a needed construct. *Entrep. Theory Pract.* **2011**, *35*, 855–872. [CrossRef]
112. Engelen, A.; Gupta, V.; Strenger, L.; Brettel, M. Entrepreneurial orientation, firm performance, and the moderating role of transformational leadership behaviors. *J. Manag.* **2015**, *41*, 1069–1097. [CrossRef]
113. Davis, J.L.; Bell, R.G.; Payne, G.T.; Kreiser, P.M. Entrepreneurial orientation and firm performance: The moderating role of managerial power. *Am. J. Bus.* **2010**, *25*, 41–54. [CrossRef]
114. Tang, Z.; Kreiser, P.M.; Marino, L.; Dickson, P.; Weaver, K.M. A hierarchical perspective of the dimensions of entrepreneurial orientation. *Int. Entrep. Manag. J.* **2008**, *5*, 181–201. [CrossRef]
115. Pratono, A.H.; Mahmood, R. Entrepreneurial orientation and firm performance: How can micro, small and medium-sized enterprises survive environmental turbulence? *Pac. Sci. Rev. B Humanit. Soc. Sci.* **2015**, *1*, 85–91. [CrossRef]
116. Rajapathirana, R.J.; Hui, Y. Relationship between innovation capability, innovation type, and firm performance. *J. Innov. Know.* **2018**, *3*, 44–55. [CrossRef]
117. Bain, D.; Kleinknecht, A. *New Concepts in Innovation Output Measurement*; Springer: Berlin/Heidelberg, Germany, 2016.
118. Cruz, C.; Nordqvist, M. Entrepreneurial orientation in family firms: A generational perspective. *Small Bus. Econ.* **2012**, *38*, 33–49. [CrossRef]
119. Arteaga, R.; Escribá-Esteve, A. Heterogeneity in family firms: Contextualising the adoption of family governance mechanisms. *J. Fam. Bus. Manag.* **2020**, *11*, 200–222. [CrossRef]
120. Jaskiewicz, P.; Dyer, W.G. Addressing the elephant in the room: Disentangling family heterogeneity to advance family business research. *Fam. Bus. Rev.* **2017**, *30*, 111–118. [CrossRef]
121. De Massis, A.; Wang, H.; Chua, J.H. Counterpoint: How heterogeneity among family firms influences organizational change. *J. Chang. Manag.* **2019**, *19*, 37–44. [CrossRef]
122. Núñez-Pomar, J.; Prado-Gascó, V.; Sanz, V.A.; Hervás, J.C.; Moreno, F.C. Does size matter? Entrepreneurial orientation and performance in Spanish sports firms. *J. Bus. Res.* **2016**, *69*, 5336–5341. [CrossRef]
123. Bornhausen, A.M. Conceptualizing cross-country analyses of family firms: A systematic review and future research agenda. *Int. Bus. Rev.* **2022**, *31*, 101924. [CrossRef]
124. Bratnicka-Myśliwiec, K.; Ingram, T. Demystifying the role of slack resources and paradox mindset for organizational creativity in family and non-family firms. *Manag. Issues* **2022**, *20*, 176–196. [CrossRef]
125. Rezaei, J.; Ortt, R. Entrepreneurial orientation and firm performance: The mediating role of functional performances. *Manag. Res. Rev.* **2018**, *41*, 878–900. [CrossRef]
126. Escamilla-Fajardo, P.; Núñez-Pomar, J.M.; Prado-Gascó, V. Economic performance in Spanish sports clubs: Entrepreneurial orientation of professional and non-professional teams analysed through fsQCA. *Eur. J. Int. Manag.* **2021**, *15*, 214–230. [CrossRef]
127. Vaillant, Y.; Lafuente, E. Entrepreneurial experience and the innovativeness of serial entrepreneurs. *Manag. Dec.* **2018**, *57*, 2869–2889. [CrossRef]
128. Francioni, B.; Musso, F.; Cioppi, M. Decision-maker characteristics and international decisions for SMEs. *Manag. Dec.* **2015**, *53*, 2226–2249. [CrossRef]

129. Saiz-Álvarez, J.M.; Cuervo-Arango, C.; Coduras, A. Entrepreneurial strategy, innovation, and cognitive capabilities: What role for intuitive SMEs? *J. Small Bus. Strategy* **2013**, *23*, 29–40.
130. Bourkha, B.; Belfellah, Y. Inter-organizational imitation: Definition and typology. *Acc. Fin. Control* **2017**, *1*, 23–31. [CrossRef]
131. Freiling, J.; Lütke Schelhowe, C. The impact of entrepreneurial orientation on the performance and speed of internationalization. *J. Entrep. Manag. Innov.* **2014**, *10*, 169–199. [CrossRef] [PubMed]
132. Shouyu, C. The relationship between innovation and firm performance: A literature review. In Proceedings of the 2017 7th International Conference on Social Network, Communication and Education (SNCE 2017), Shenyang, China, 28–30 July 2017; Atlantis Press: Dordrecht, The Netherlands, 2017; pp. 648–652.
133. Chenhall, R.H.; Langfield-Smith, K. Multiple perspectives of performance measures. *Eur. Manag. J.* **2007**, *25*, 266–282. [CrossRef]
134. Anwar, M.; Shah, S.Z. Entrepreneurial orientation and generic competitive strategies for emerging SMEs: Financial and nonfinancial performance perspective. *J. Public Aff.* **2020**, *21*, 1–17. [CrossRef]
135. Combs, J.G.; Crook, T.R.; Shook, C.L. The Dimensionality of Organizational Performance and its Implications for Strategic Management Research. *Res. Methodol. Strategy Manag.* **2005**, *2*, 259–286.
136. Al-Mamary, Y.H.; Alwaheeb, M.A.; Alshammari, N.G.; Abdulrab, M.; Balhareth, H.; Soltane, H.B. The effect of entrepreneurial orientation on financial and non-financial performance in Saudi SMEs: A review. *J. Crit. Rev.* **2020**, *7*, 200–208.
137. Lumpkin, G.T. From legitimacy to impact: Moving the field forward by asking how entrepreneurship informs life. *Strateg. Entrep. J.* **2011**, *5*, 3–9. [CrossRef]